Fiscal Year:	FY 2019	**Task Last Updated:**	FY 05/02/2019
PI Name:	Nicholson, Wayne Ph.D.		
Project Title:	Comparative Evaluation of Microbial Transcriptomic Responses to Spaceflight Stress: Elucidating Underlying Molecular Mechanisms		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Comments:	NOTE: Congressional District is for University of Florida (32611-0001)--Ed., 8/17/2012; changed to 3 (Ed., 7/30/13)		
Project Type:	Ground,NASA GeneLab	**Solicitation / Funding Source:**	2016 Space Biology (ROSBio) NNH16ZTT001N-GeneLab. Appendix A: Translational Systems Biology and Informatics Research Using the GeneLab Data System
Start Date:	12/19/2016	**End Date:**	12/18/2019
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA KSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/18/2019 per NSSC information (Ed., 1/11/2019)		
Key Personnel Changes/Previous PI:	October 2017 report: None. October 2018 report: None		
COI Name (Institution):			
Grant/Contract No.:	NNX17AD51G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Microbial cells exposed to the human spaceflight environment are observed to respond differently to similarly-treated ground controls, a phenomenon dubbed the “spaceflight syndrome.” Over the past 4 decades, a number of experiments have been conducted inside various vehicles (e.g., Shuttle, Mir, ISS (International Space Station)) to understand how microbes sense and respond to stresses of human spaceflight including, but not limited to, microgravity and ionizing radiation. In early studies, various phenotypic outputs were measured, such as: growth rate and yield; virulence; biofilm formation and architecture; and resistance to antibiotics or abiotic stresses. With the advent of the genomics and post-genomics revolutions in biology, there have been initial efforts to understand more fundamental molecular aspects of the spaceflight syndrome, by performing global-scale "-omics" analyses of the transcriptome, proteome, metabolome, etc. of microbes exposed to spaceflight. Such studies have yielded valuable insights into the molecular responses of certain individual microbes to the spaceflight environment; furthermore, datasets from these experiments have been deposited into a common, publicly available database, the GeneLab Data System (GDS). We propose to use the data in the GDS as a starting point to explore commonalities and differences in the responses to spaceflight of various different microbes, with an eye towards uncovering possible common fundamental mechanisms. Working Hypothesis: Exposure to the human spaceflight environment alters the physicochemical environment immediately surrounding bacterial cells and within the cells themselves (mainly due to microgravity effects and ionizing radiation). Cells sense the altered conditions and respond using the systems they possess--which evolved on Earth, not in space. A large proportion of microbial physiological responses are regulated at the level of transcription, thus the transcriptome is a logical place to search for common fundamental mechanisms. Objectives: In order to address this hypothesis, we propose the following Specific Aims: 1. Mine the GDS bacterial transcriptome database and reprocess the data in a fashion that will allow cross-organism comparisons. 2. Search the spaceflight literature to identify additional relevant datasets that may have escaped inclusion in the GDS. Deposit these additional datasets in the GDS. Access and reformat the additional data as in Specific Aim 1. 3. Perform bioinformatics analyses to identify pathways and regulons significantly activated or repressed by exposure to spaceflight vs. ground controls in individual organisms. 4. Compare transcriptomes of the same organism grown in simulated microgravity (clinostats) vs. actual spaceflight conditions. 5. Perform comparisons across species, genera, families, and higher taxonomic groupings to find similarities and differences in spaceflight responses. 6. Develop specific hypotheses from the comparisons in Specific Aim 5 and perform wet-lab experiments to test these hypotheses in ground-based systems and in future spaceflight missions. Expected significance: Microbes are ubiquitous in space vehicles, as well as on and within humans themselves. Because microbes play such important and essential roles in the health and well-being of astronauts and their life-support systems, it is imperative to understand their responses to the human spaceflight environment. The analyses proposed here will attempt to systematically identify common molecular mechanisms used by a wide range of bacteria to cope with the stresses encountered in human spaceflight, a primary goal of both the GeneLab endeavor and the broader mission of the NASA Space Biology Program. In addition, the pipelines and protocols developed in this study will have applications beyond GeneLab, and in principle can be applied to comparison of the transcriptomic responses of any collection of microbes to any given type of stress.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Microbes are ubiquitous in space vehicles, as well as on and within humans themselves. Because microorganisms play such important and essential roles in the health and well-being of astronauts and their life-support systems, it is imperative to understand their responses to the human spaceflight environment. The analyses performed in this study will systematically identify common molecular mechanisms used by a wide range of bacteria to cope with the stresses encountered in human spaceflight--a primary goal of both the GeneLab endeavor and the broader mission of the NASA Space Biology Program. In addition, the pipelines and protocols developed in this study will have applications beyond GeneLab, and in principle can be applied to comparison of the transcriptomic responses of any collection of microbes to any given type of stress.
Task Progress:	We had previously conducted 2 spaceflight experiments Biological Research in Canisters (BRIC) (BRIC-21 and BRIC-23) in which we measured the transcriptomic response of Bacillus subtilis cells to spaceflight. Graduate student Michael Morrison tested the hypothesis that growth of B. subtilis under comparable conditions in a clinostat could replicate the spaceflight results, using his bioinformatics pipeline and statistical analyses. He found almost no correlation between the clinostat results with the spaceflight results for B. subtilis. Subsequently, Michael mined the GeneLab Data System to identify all clinostat transcriptome data sets for several different microbes to compare with their corresponding flight experiments, and again found a very low degree of correlation between the clinostat and the spaceflight data. These results have been presented at recent scientific conferences. A manuscript describing our results has been submitted to Nature Scientific Reports for review.
Bibliography Type:	Description: (Last Updated: 07/25/2025)
Abstracts for Journals and Proceedings	Morrison MD, Fajardo-Cavazos P, Nicholson WL. "Comparison of Bacillus subtilis transcription profiles from separate missions to the ISS reveals common responses." Talk presented at the 2018 Astrobiology Graduate Conference, Atlanta, GA, June 4-8, 2018. 2018 Astrobiology Graduate Conference, Atlanta, GA, June 4-8, 2018. Abstracts, Session VI-4. , Jun-2018

Abstracts for Journals and Proceedings	Morrison MD, Fajardo-Cavazos P, Nicholson WL. "Global RNA-seq analyses reveal a dramatic lack of concordance between Bacillus subtilis transcriptomes from spaceflight- vs. clinostat-grown cultures." Talk presented at the Committee on Space Research (COSPAR) 2018 42nd Scientific Assembly, Pasadena, CA, July 14-22, 2018. Committee on Space Research (COSPAR) 2018 42nd Scientific Assembly, Pasadena, CA, July 14-22, 2018. Abstract F2.4-0003-18. , Jul-2018
Abstracts for Journals and Proceedings	Morrison MD, Fajardo-Cavazos P, Nicholson WL. "Lack of concordance between Bacillus subtilis spaceflight and clinostat transcription profiles." Talk presented at the 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. Abstract #93. , Oct-2018
Articles in Peer-reviewed Journals	Morrison MD, Nicholson WL. "Meta-analysis of data from spaceflight transcriptome experiments does not support the idea of a common bacterial “spaceflight response.” " Sci Rep. 2018 Sep 26;8(1):14403. https://doi.org/10.1038/s41598-018-32818-z ; PubMed PMID: 30258082; PubMed Central PMCID: PMC6158273 , Sep-2018
Articles in Peer-reviewed Journals	Morrison MD, Fajardo-Cavazos P, Nicholson WL. "Comparison of Bacillus subtilis transcriptome profiles from two separate missions to the International Space Station." npj Microgravity 2019 Jan;5:1. eCollection 2019. https://doi.org/10.1038/s41526-018-0061-0 ; PubMed PMID: 30623021; PubMed Central PMCID: PMC6323116 , Jan-2019